

BUDGET VS ACTUAL FINANCIAL ANALYSIS

Complete Project Documentation – Excel + Power BI

Cleaned dataset • Pivot Tables • Excel Dashboard • Power BI Data Model • DAX • Visualizations • Validation • Gap Analysis

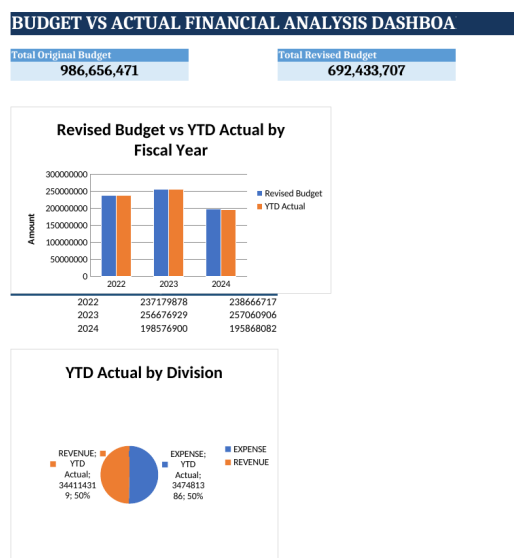


Figure 1. Source Excel dashboard screenshot.

Project item	Status
Cleaned Excel workbook	Completed
Pivot Table analysis	Completed
Excel dashboard	Completed
Power BI PBIX model	Available / inspected
Power BI visuals	Available / inspected
Documentation	Expanded and gap-checked

1. Executive Summary

The final workbook contains 1,061 cleaned records and 11 fields. The dataset covers fiscal years 2022–2024 and includes Original Budget, Revised Budget and YTD Actual. The final workbook contains the sheets Original Data, Cleaned Data, Pivot Tables, Dash Board and Project notes.

At the portfolio level, Original Budget totals 986,656,471, Revised Budget totals 692,433,707, and YTD Actual totals 691,595,705. Using the final workbook's stated convention (YTD Actual – Revised Budget), the total variance is -838,002 and utilisation is 99.9%.

Key management takeaways

- 2023 has the highest YTD Actual total at 257,060,906.
- TRANSIT is the largest Function by YTD Actual at 345,432,934.
- TRANSIT is also the largest Department by YTD Actual at 399,794,829.
- GENERAL FUND is the largest Fund by YTD Actual at 401,683,631.
- Expense YTD Actual is 347,481,386; Revenue YTD Actual is 344,114,319.

2. Project Overview and Objective

This project demonstrates an end-to-end financial analysis workflow: source data is cleaned and transformed in Excel, summarised with Pivot Tables, presented through an Excel dashboard, and then modelled and visualised in Power BI.

Objectives

- Validate and standardise the financial dataset.
- Handle missing Fiscal Year, Account Type and YTD Actual fields according to the documented project logic.
- Create budget-versus-actual variance and utilisation metrics.
- Summarise performance by fiscal year, division, function, fund and department.
- Build a reusable Power BI fact/dimension model and report visuals.
- Provide clear documentation suitable for project submission and GitHub presentation.

Project workflow

Stage	Output
Raw / original data	Original Data sheet
Cleaning & transformation	Cleaned Data sheet
Summarisation	Pivot Tables sheet
Reporting	Dash Board sheet
Documentation of actions	Project notes sheet
BI modelling	Power BI Fact_Budget + dimensions
BI reporting	Power BI report pages and visuals

3. Data Sources and Workbook Structure

Source / Sheet	Purpose	Evidence
Original Data	Starting financial dataset	Supplied workbook
Cleaned Data	Final standardised analysis table	1,061 rows × 11 fields
Pivot Tables	Fiscal year, division, function, fund and department summaries	Supplied workbook
Dash Board	KPI and chart presentation	Supplied workbook
Project notes	Cleaning, validation, transformation and dashboard log	Supplied workbook
Power BI PBIX	Data model, DAX/report visual structure	Supplied PBIX

OBJECTID	Fiscal Year	Fund	Function
26	2023	GENERAL FUND	LEISURE
36	2024	GRANTS FUND	PUBLIC SAFETY
37	2022	DOWNTOWN SERVICE	ENVIRONMENT & DEV
45	2024	CAPITAL IMPROVEME	CAPITAL
51	2024	DOWNTOWN SERVICE	ENVIRONMENT & DEV
52	2022	CAPITAL IMPROVEME	CAPITAL
59	2023	GENERAL FUND	LEISURE
61	2022	CAPITAL IMPROVEME	CAPITAL
71	2023	STORMWATER MANA	STORMWATER
83	2023	TRANSIT FUND	TRANSIT
108	2024	OFF-STREET PARKING	PARKING
110	2023	CAPITAL IMPROVEME	CAPITAL
111	2024	GENERAL FUND	GENERAL GOVERNME
119	2023	GRANTS FUND	PUBLIC SAFETY
120	2024	GRANTS FUND	PUBLIC SAFETY
121	2022	CAPITAL IMPROVEME	CAPITAL
123	2024	GENERAL FUND	LEISURE
133	2022	GENERAL FUND	LEISURE
143	2023	GENERAL FUND	GENERAL GOVERNME
152	2023	TRANSIT FUND	TRANSIT
155	2023	OFF-STREET PARKING	PARKING
156	2024	ON-STREET PARKING	PARKING
161	2023	TRANSIT FUND	TRANSIT
171	2024	CAPITAL IMPROVEME	CAPITAL
178	2022	TRANSIT FUND	TRANSIT
182	2023	AFFORD HSG DEVELOP	ENVIRONMENT & DEV
197	2023	DOWNTOWN SERVICE	ENVIRONMENT & DEV
204	2024	GRANTS FUND	PUBLIC SAFETY
218	2023	VEHICLE MAINTENAN	INTERNAL SERVICES
221	2023	GRANTS FUND	PUBLIC SAFETY
222	2024	ON-STREET PARKING	PARKING
234	2024	TRANSIT FUND	TRANSIT
240	2024	CAPITAL RESERVE FUN	CAPITAL
245	2023	TRANSIT FUND	TRANSIT
250	2022	GRANTS FUND	PUBLIC SAFETY
253	2022	VEHICLE REPLACEME	INTERNAL SERVICES
267	2024	LIBRARY GIFT FUND	LEISURE
271	2022	COMPUTER REPLACEM	INTERNAL SERVICES
272	2023	CAPITAL IMPROVEME	CAPITAL
277	2022	OFF-STREET PARKING	PARKING
278	2023	TRANSIT FUND	TRANSIT
284	2023	VEHICLE MAINTENAN	INTERNAL SERVICES
285	2024	ON-STREET PARKING	PARKING
292	2022	TRANSIT FUND	TRANSIT
293	2023	CAPITAL IMPROVEME	CAPITAL
294	2024	LIBRARY GIFT FUND	LEISURE
302	2023	LIBRARY GIFT FUND	LEISURE
309	2024	TRANSIT FUND	TRANSIT
310	2022	STORMWATER MANA	STORMWATER
318	2024	VEHICLE MAINTENAN	INTERNAL SERVICES

Figure 2. Source workbook data structure screenshot.

4. Attribute / Column Details

Attribute	Type	Role / Description
OBJECTID	Integer	Unique record identifier used for duplicate checking.
Fiscal Year	Integer	Fiscal year; final workbook covers 2022, 2023 and 2024.
Fund	Text	Fund classification.
Function	Text	Functional area such as TRANSIT, PUBLIC SAFETY and CAPITAL.
Department	Text	Operational department.
Division	Text	REVENUE or EXPENSE division.
Account Type	Text	Revenue or Expense classification derived from Division.
Category	Text / code	Category code retained as text.
Original Budget	Numeric	Original budget amount.
Revised Budget	Numeric	Revised budget amount used for variance/utilisation.
YTD Actual	Numeric	Year-to-date actual amount used in analysis.

Department	Division	Account Type	Category
LIBRARY	EXPENSE	Expense	Unknown
POLICE	EXPENSE	Expense	Unknown
MANAGER	REVENUE	Revenue	Unknown
FIRE	REVENUE	Revenue	Unknown
MANAGER	REVENUE	Revenue	Unknown
TRANSIT	REVENUE	Revenue	Unknown
PARKS & RECREATION	EXPENSE	Expense	Unknown
PUBLIC WORKS	REVENUE	Revenue	Unknown
STORMWATER	REVENUE	Revenue	Unknown
TRANSIT	EXPENSE	Expense	Unknown
PARKING	EXPENSE	Expense	Unknown
PARKS & RECREATION	EXPENSE	Expense	Unknown
COMM & PUBLIC AFF	EXPENSE	Expense	Unknown
POLICE	REVENUE	Revenue	Unknown
POLICE	REVENUE	Revenue	Unknown
TRANSIT	REVENUE	Revenue	Unknown
LIBRARY	EXPENSE	Expense	Unknown
LIBRARY	EXPENSE	Expense	Unknown
COMM & PUBLIC AFF	EXPENSE	Expense	Unknown
TRANSIT	EXPENSE	Expense	Unknown
PARKING	EXPENSE	Expense	Unknown
PARKING	EXPENSE	Expense	Unknown
TRANSIT	EXPENSE	Expense	Unknown
PARKS & RECREATION	EXPENSE	Expense	Unknown
TRANSIT	EXPENSE	Expense	Unknown
HOUSING & COMMUN	REVENUE	Revenue	Unknown
MANAGER	REVENUE	Revenue	Unknown
POLICE	EXPENSE	Expense	Unknown
PUBLIC WORKS	REVENUE	Revenue	Unknown
POLICE	REVENUE	Revenue	Unknown
PARKING	REVENUE	Revenue	Unknown
TRANSIT	EXPENSE	Expense	Unknown
TRANSIT	REVENUE	Revenue	Unknown
TRANSIT	EXPENSE	Expense	Unknown
POLICE	REVENUE	Revenue	Unknown
PUBLIC WORKS	EXPENSE	Expense	Unknown
LIBRARY	REVENUE	Revenue	Unknown
TECHNOLOGY SOLUTI	REVENUE	Revenue	Unknown
PARKS & RECREATION	EXPENSE	Expense	Unknown
PARKING	EXPENSE	Expense	Unknown
TRANSIT	EXPENSE	Expense	Unknown
PUBLIC WORKS	REVENUE	Revenue	Unknown
PARKING	EXPENSE	Expense	Unknown
TRANSIT	EXPENSE	Expense	Unknown
TRANSIT	REVENUE	Revenue	Unknown
LIBRARY	EXPENSE	Expense	Unknown
LIBRARY	EXPENSE	Expense	Unknown
TRANSIT	REVENUE	Revenue	Unknown
STORMWATER	EXPENSE	Expense	Unknown
PUBLIC WORKS	REVENUE	Revenue	Unknown

Figure 3. Source screenshot showing Department, Division, Account Type and Category after transformation.

5. Data Pre-Processing and Cleaning

The final workbook's Project notes document the following cleaning and transformation actions.

Task	Observation / Issue	Final action
Duplicate records	Potential duplicate records	OBJECTID checked; no duplicate IDs found.
Fiscal Year	Missing in the earlier cleaned part	Filled using the documented 2022/2023/2024 repeating pattern.
Account Type	Missing in the earlier cleaned part	Derived from Division: REVENUE → Revenue; other records → Expense.
YTD Actual	Missing in the earlier cleaned part	Calculated from Revised Budget using the documented 95% / 105% logic.
Category	Formatting / type consistency	Standardised as text to preserve category codes.
Budget fields	Numeric financial values	Original Budget and Revised Budget retained as numeric amounts.

Quality check after final cleaning

- Rows checked: 1,061
- Duplicate OBJECTID count: 0
- Blank cells across cleaned data: 0
- Fiscal Year values present for all rows.
- Account Type values present for all rows.
- YTD Actual values present for all rows.

Project Area	Observation / Task
Data Cleaning	Duplicate records
Data Cleaning	Fiscal Year was missing
Data Cleaning	Account Type was missing
Data Cleaning	YTD Actual was missing
Data Cleaning	Category formatting
Data Validation	Budget fields
Transformation	Variance
Transformation	Utilization
Pivot Analysis	Fiscal Year
Pivot Analysis	Division
Pivot Analysis	Function
Pivot Analysis	Fund
Pivot Analysis	Department
Dashboard	KPI cards
Dashboard	Charts
Deliverable	Final workbook

Figure 4. Source Project notes screenshot – cleaning observations/tasks.

6. Cleaning Actions – Evidence

Action Taken / Output

Checked OBJECTID; no duplicate IDs were found.

Filled Fiscal Year using the 2022, 2023, 2024 repeating pattern from the cleaned Part 1 logic.

Derived Account Type from Division: REVENUE → Revenue; other records → Expense.

Calculated YTD Actual from Revised Budget using the alternating 95% / 105% logic in the cleaned Part 1 formulas.

Standardized Category values as text to preserve category codes.

Original Budget and Revised Budget retained as numeric financial amounts.

Calculated YTD Actual – Revised Budget for analysis.

Calculated YTD Actual ÷ Revised Budget for dashboard KPI analysis.

Budget and actual totals summarized by fiscal year.

Budget and actual totals summarized by Revenue/Expense division.

Functions ranked by YTD Actual.

Budget and actual totals summarized by Fund.

Top 10 departments identified by YTD Actual.

Total Original Budget, Total Revised Budget, Total YTD Actual, Budget Variance and Utilization %.

Fiscal Year comparison, Top Functions, YTD Actual by Division and Top 10 Departments.

Includes Original Data, Cleaned Data, Pivot Tables, Dash Board and Project notes.

Figure 5. Source Project notes screenshot – actions taken and outputs.

7. Pivot Table Analysis

The Pivot Tables sheet summarises Original Budget, Revised Budget, YTD Actual and Variance by fiscal year, division, function, fund and department. The final workbook uses $\text{Variance} = \text{YTD Actual} - \text{Revised Budget}$.

Financial Analysis - Pivot Tables				
By Fiscal Year				
Fiscal Year	Original Budget	Revised Budget	YTD Actual	
2,022	397,375,296	237,179,878	238,666,717	
2,023	318,451,949	256,676,929	257,060,906	
2,024	270,829,226	198,576,900	195,868,082	

By Function				
Function	Original Budget	Revised Budget	YTD Actual	
TRANSIT	485,221,840	346,673,419	345,432,934	
PUBLIC SAFETY	92,603,794	70,967,563	70,508,442	
ENVIRONMENT & DEV	87,980,707	63,073,275	62,539,078	
GENERAL GOVERNME	68,367,841	51,644,145	51,471,044	
DEBT SERVICE	61,895,426	45,742,572	47,184,949	
LEISURE	45,177,564	35,188,205	35,339,819	
PARKING	25,956,063	18,703,516	18,819,089	
INTERNAL SERVICES	29,492,650	17,554,254	17,413,380	
STORMWATER	41,989,070	16,269,341	16,567,413	
PUBLIC HOUSING	19,145,065	13,351,299	13,475,411	
CAPITAL	28,826,451	13,266,118	12,844,146	

Figure 6. Pivot Table screenshot – Fiscal Year and Function analysis.

7.1 Pivot Analysis – Division and Fund



Original Budget	Revised Budget	YTD Actual	Variance
492,000,105	348,985,886	347,481,386	-1,504,500
494,656,366	343,447,821	344,114,319	666,498

Original Budget	Revised Budget	YTD Actual	Variance
545,477,514	402,606,997	401,683,631	-923,366
206,081,596	150,533,464	149,025,919	-1,507,545
61,895,426	45,742,572	47,184,949	1,442,377
41,989,070	16,269,341	16,567,413	298,072
19,145,065	13,351,299	13,475,411	124,112
18,595,623	13,261,029	13,328,654	67,625
28,826,451	13,264,673	12,842,682	-421,991
11,637,154	9,147,100	8,943,301	-203,799
16,639,544	7,578,470	7,610,977	32,507
7,360,440	5,442,487	5,490,435	47,948
9,825,624	4,074,197	4,162,326	88,129
6,449,482	3,120,251	3,212,899	92,648
4,939,658	2,809,862	2,770,385	-39,477
3,105,878	2,449,834	2,487,627	37,793
3,471,994	1,952,002	1,948,530	-3,472
1,215,952	828,684	859,102	30,418
0	1,445	1,464	19

Figure 7. Pivot Table screenshot – Division and Fund analysis.

7.2 Pivot Analysis – Top Departments



Top 10 Departmente	
Department	Revised Budget
TRANSIT	399,890,903
PUBLIC WORKS	58,500,797
POLICE	46,266,677
FIRE	32,099,118
PARKS & RECREATION	23,765,285
NON-DEPARTMENTAL	23,029,745
PARKING	18,703,516
STORMWATER	16,269,341
PUBLIC HOUSING	13,458,903
LIBRARY	12,475,302

Figure 8. Pivot Table screenshot – Top 10 Departments by Revised Budget.

8. KPI Definitions and Business Logic

KPI / Metric	Definition	Purpose
Total Original Budget	SUM(Original Budget)	Baseline budget comparison.
Total Revised Budget	SUM(Revised Budget)	Current revised budget position.
Total YTD Actual	SUM(YTD Actual)	Actual financial position.
Budget Variance	YTD Actual – Revised Budget	Positive = actual exceeds revised budget; negative = actual is below revised budget.
Utilization %	YTD Actual ÷ Revised Budget	Measures actual relative to revised budget.
Budget Status	Compare YTD Actual and Revised Budget	Classifies rows as over / equal / under budget.

Portfolio KPI values from the final cleaned dataset

Metric	Value
Total Original Budget	986,656,471
Total Revised Budget	692,433,707
Total YTD Actual	691,595,705
Total Variance (YTD – Revised)	-838,002
Utilization	99.9%
Rows over revised budget	454
Rows equal to revised budget	142
Rows below revised budget	465

9. Excel Dashboard – Visual Documentation

The Excel dashboard combines KPI cards, fiscal-year comparison, division mix, top functions and department summaries. The following screenshots are the source workbook dashboard pages.

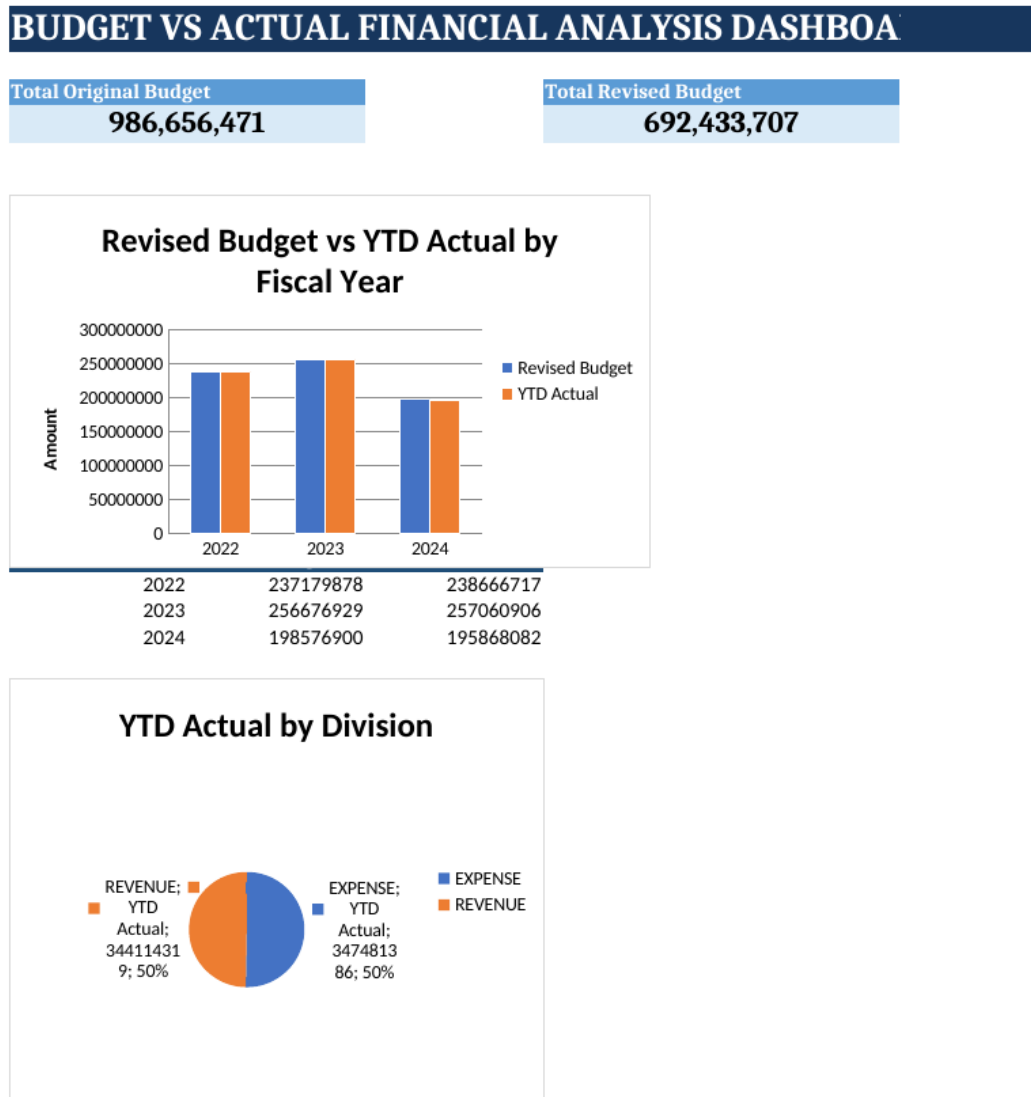


Figure 9. Excel dashboard – KPI cards, fiscal-year comparison and YTD Actual by Division.

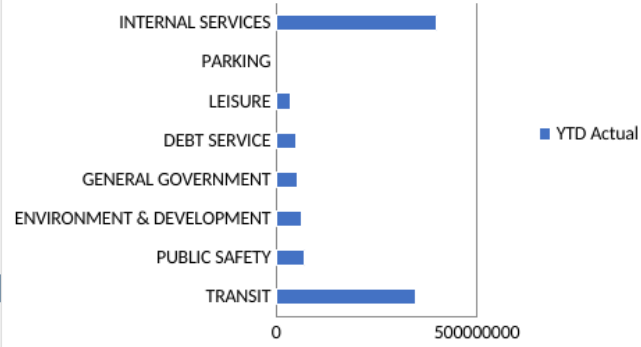
Total YTD Actual

691,595,705

Budget Variance

-838

Top Functions by YTD Actual



Function

TRANSIT

PUBLIC SAFETY

ENVIRONMENT & DE

GENERAL GOVERNMENT

DEBT SERVICE

LEISURE

PARKING

INTERNAL SERVICES

Top 10 Department

Department	Revised Budget	YTD Actual	Variance
TRANSIT	399,890,903	399,794,829	-96,074
PUBLIC WORKS	58,500,797	57,801,935	-698,862
POLICE	46,266,677	45,818,539	-448,138
FIRE	32,099,118	31,996,530	-102,588
PARKS & RECREATIO	23,765,285	23,936,711	171,426
NON-DEPARTMENTA	23,029,745	22,777,189	-252,556
PARKING	18,703,516	18,819,089	115,573
STORMWATER	16,269,341	16,567,413	298,072
PUBLIC HOUSING	13,458,903	13,586,623	127,720
LIBRARY	12,475,302	12,449,501	-25,801

Figure 10. Excel dashboard – Total YTD Actual, Budget Variance, Top Functions and Top Departments.

10. Power BI Data Modelling

The supplied PBIX report definition contains a Fact_Budget table and dimension tables for Account Type, Category, Department, Fiscal Year, Function and Fund. This is a star-schema style design in which the fact table stores financial records and dimensions provide filtering/grouping attributes.

Table	Type	Main role
Fact_Budget	Fact	Financial records and numeric budget/actual fields.
Dim_FiscalYear	Dimension	Fiscal year filtering.
Dim_Fund	Dimension	Fund filtering.
Dim_Function	Dimension	Function filtering.
Dim_Department	Dimension	Department filtering.
Dim_Category	Dimension	Category filtering.
Dim_AccountType	Dimension	Revenue / Expense filtering.

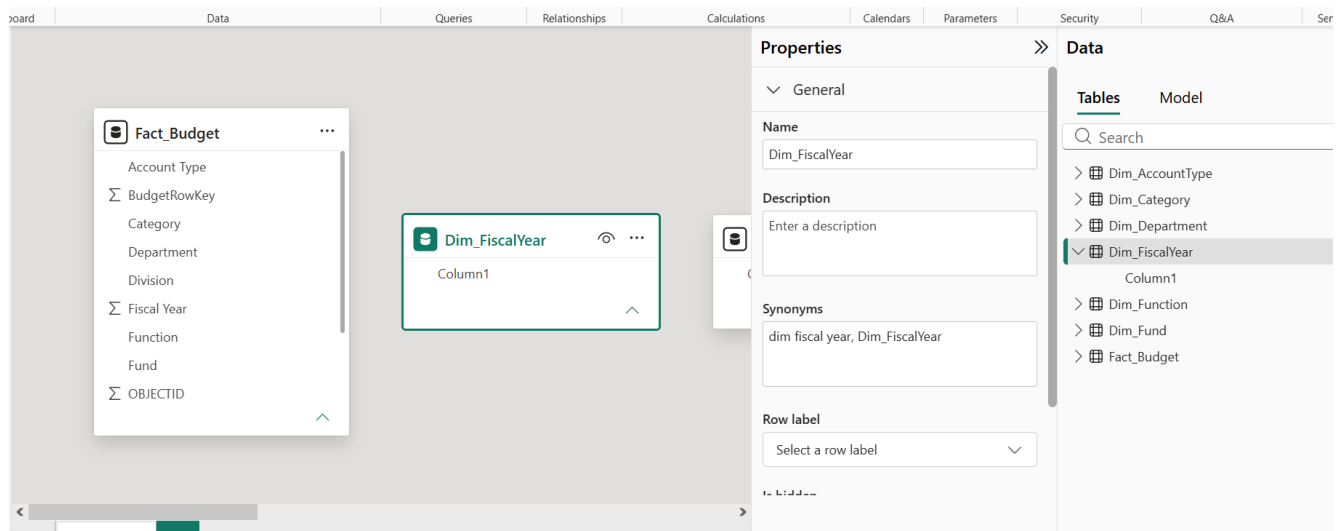


Figure 12. User-provided Power BI model screenshot showing Fact_Budget and dimension tables.

11. Relationships – Validation and Documentation

A user-provided screenshot records the relationship setup attempt between Fact_Budget[Fiscal Year] and Dim_Fiscal year[Fiscal Year]. The dialog shows Many-to-one (*:1), Single cross-filter direction and an active relationship. It also shows a null value in the dimension table preview, which is a data-model quality concern because a clean dimension key should be unique and complete.

From table

Fact_Budget

Account Type	BudgetRowKey	Category	Department	Division	Fiscal Year	Function
Expense	6	1188389	TRANSIT	EXPENSE	2024	DEBT S
Revenue	27	1359731	TRANSIT	REVENUE	2024	DEBT S
Expense	40	20000	TRANSIT	EXPENSE	2022	TRANS

To table

Dim_Fiscal year

Fiscal Year
null
2022
2023

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

Save

Cancel

Figure 13. User-provided Power BI relationship setup screenshot.

Recommended relationship rule for the final documentation

- Fact_Budget[Fiscal Year] → Dim_FiscalYear[Fiscal Year]: many-to-one, single direction, active.
- Dimension key must contain one row per fiscal year with no duplicate key values.
- Apply the same one-to-many pattern for Fund, Function, Department, Category and Account Type.
- Validate relationship status after all dimensions are loaded.

12. Power BI DAX Documentation

The project documentation requires DAX calculations for key metrics. The formulas below are the clean, auditable versions that match the final workbook's business logic.

Measure	DAX expression	Use
Total Original Budget	SUM(Fact_Budget[Original Budget])	Total original budget.
Total Revised Budget	SUM(Fact_Budget[Revised Budget])	Total revised budget.
Total Actual	SUM(Fact_Budget[YTD Actual])	Total YTD actual.
Budget Variance	[Total Actual] - [Total Revised Budget]	YTD Actual minus Revised Budget.
Utilization %	DIVIDE([Total Actual], [Total Revised Budget], 0)	Actual as % of revised budget.
Budget Status	IF([Total Actual] > [Total Revised Budget], "Over Budget", IF([Total Actual] < [Total Revised Budget], "Under Budget", "On Budget"))	Budget Status: "Over Budget", "Under Budget", "On Budget"

Important consistency fix

The earlier documentation version used Revised Budget – YTD Actual for variance. The final workbook's Project notes explicitly state YTD Actual – Revised Budget, and the Pivot Tables show the same convention. This expanded document uses the final workbook convention throughout.

13. Power BI Report Pages and Visualizations

The supplied PBIX contains 7 report pages. Its visual definitions include cards, clustered column charts, bar charts, a donut chart and slicers. The page map below documents the structure found in the PBIX.

Power BI Report Page Map

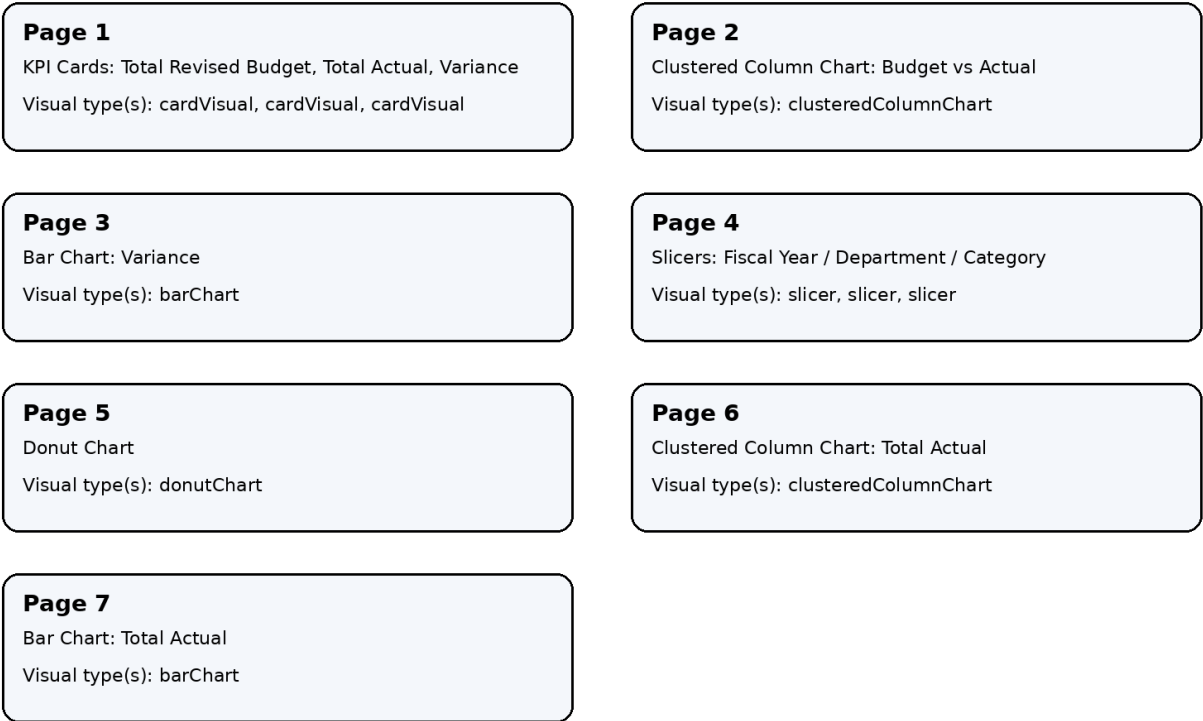


Figure 14. Power BI report page map reconstructed from the supplied PBIX report definition.

14. Visualization Catalogue

Visual	Dimension / Axis	Measure / Value	Business question
KPI Card	None	Total Revised Budget	What is the revised budget?
KPI Card	None	Total Actual	How much has been recorded as actual?
KPI Card	None	Variance	Is actual above or below revised budget?
Clustered Column	Fiscal Year	Total Revised Budget vs Total Actual	How does performance change by year?
Bar Chart	Department / Function	Variance or Total Actual	Which areas drive the largest differences?
Donut	Budget Status	Count / value	How are records distributed by budget status?
Slicer	Fiscal Year	Filter	How does the selected year affect the report?
Slicer	Department	Filter	How does a selected department perform?
Slicer	Category	Filter	How does category selection affect the analysis?

Recommended interaction standards

- Use slicers for Fiscal Year, Department and Category.
- Use clear visual titles and data labels.
- Keep KPI cards at the top of the report.
- Use a consistent sign convention for variance across all pages.
- Validate that filters propagate from dimensions to Fact_Budget.

15. Analytical Results

Fiscal year summary

Fiscal Year	Original Budget	Revised Budget	YTD Actual	Variance
2022	397,375,296	237,179,878	238,666,717	1,486,839
2023	318,451,949	256,676,929	257,060,906	383,977
2024	270,829,226	198,576,900	195,868,082	-2,708,818

Division summary

Division	Revised Budget	YTD Actual	Variance
EXPENSE	348,985,886	347,481,386	-1,504,500
REVENUE	343,447,821	344,114,319	666,498

Function ranking

Function	YTD Actual	Variance
TRANSIT	345,432,934	-1,240,485
PUBLIC SAFETY	70,508,442	-459,121
ENVIRONMENT & DEVELOPMENT	62,539,078	-534,197
GENERAL GOVERNMENT	51,471,044	-173,101
DEBT SERVICE	47,184,949	1,442,377
LEISURE	35,339,819	151,614
PARKING	18,819,089	115,573
INTERNAL SERVICES	17,413,380	-140,874
STORMWATER	16,567,413	298,072
PUBLIC HOUSING	13,475,411	124,112

16. Top Funds and Departments

Top 10 Funds by YTD Actual

Fund	YTD Actual	Variance
GENERAL FUND	401,683,631	-923,366
TRANSIT FUND	149,025,919	-1,507,545
DEBT SERVICE FUND	47,184,949	1,442,377
STORMWATER MANAGEMENT FUND	16,567,413	298,072
PUBLIC HOUSING OPERATIONS FUND	13,475,411	124,112
OFF-STREET PARKING FUND	13,328,654	67,625
CAPITAL IMPROVEMENTS FUND	12,842,682	-421,991
VEHICLE MAINTENANCE FUND	8,943,301	-203,799
VEHICLE REPLACEMENT FUND	7,610,977	32,507
ON-STREET PARKING FUND	5,490,435	47,948

Top 10 Departments by YTD Actual

Department	Revised Budget	YTD Actual	Variance
TRANSIT	399,890,903	399,794,829	-96,074
PUBLIC WORKS	58,500,797	57,801,935	-698,862
POLICE	46,266,677	45,818,539	-448,138
FIRE	32,099,118	31,996,530	-102,588
PARKS & RECREATION	23,765,285	23,936,711	171,426
NON-DEPARTMENTAL	23,029,745	22,777,189	-252,556
PARKING	18,703,516	18,819,089	115,573
STORMWATER	16,269,341	16,567,413	298,072
PUBLIC HOUSING	13,458,903	13,586,623	127,720
LIBRARY	12,475,302	12,449,501	-25,801

17. Gap Analysis – What Was Missing and How It Was Addressed

This section is added specifically to close the documentation gaps identified between the earlier project materials, the final workbook and the PBIX workflow.

Gap	Evidence	Resolution in this document
Variance sign inconsistency	Earlier documentation used Revised – Actual; final Project Description – YTD Actuals Revised.	Project Description – YTD Actuals Revised.
Relationship validation not clearly documented	Updated screenshot shows relationship setup dialog with DAX dimension relationships and key validation rules.	DAX dimension relationships and key validation rules.
Too few screenshots	Earlier PDF had a small number of reconstructed visualizations.	Added source screenshots from Pivot Tables, dashboard, project notes and Power BI.
DAX logic not fully documented	PBIX visual definitions show metric names; project notes show DAX logic.	Added DAX logic dictionary with clean expressions.
No explicit QA section	Cleaning notes record actions but not a consolidated QA checklist.	Added QA checklist, duplicate, blank-field and metric consistency checks.
Predictive analysis gap	The supplied project has no forecasting model.	Explicitly state that predictive conclusions are not supported by the supplied materials.
Dashboard interaction documentation gap	PBIX contains slicer pages.	Added slicer/interaction documentation and recommended validation rules.

18. Data Validation and Reconciliation

Final cleaned-data checks

Check	Result
Record count	1,061
Column count	11
Duplicate OBJECTID	0
Blank cells in cleaned data	0
Fiscal years present	2022, 2023, 2024
Original Budget total	986,656,471
Revised Budget total	692,433,707
YTD Actual total	691,595,705
Variance (YTD – Revised)	-838,002
Utilization	99.9%

Reconciliation rule

For every aggregate view, the same measures should reconcile back to the Cleaned Data table. For example, fiscal-year totals should sum to the portfolio totals, and division totals should reconcile to the same overall Revised Budget and YTD Actual totals.

19. Insights and Conclusions

Descriptive insights

- Total Revised Budget is 692,433,707 and Total YTD Actual is 691,595,705.
- Overall utilisation is 99.9%.
- The final dataset is distributed across fiscal years 2022–2024.

Diagnostic insights

- Overall variance is -838,002 using YTD Actual – Revised Budget.
- TRANSIT is the largest Function and Department by YTD Actual.
- The Division view is nearly balanced between Expense and Revenue in YTD Actual.

Predictive insight

- No forecasting or predictive model is supplied in the project materials; therefore a predictive conclusion is not claimed.

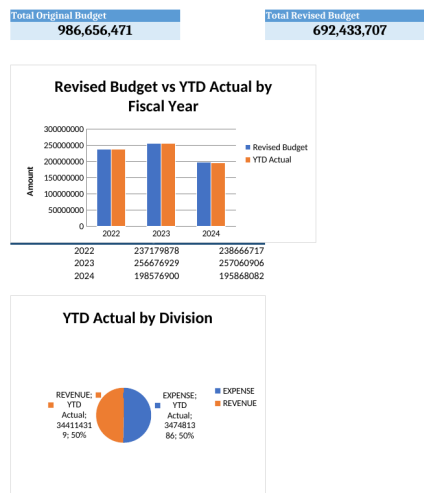
Prescriptive recommendations

- Prioritise departments/functions with the largest positive variance for budget-overflow review.
- Review high-utilisation areas close to or above 100% of revised budget.
- Use Fiscal Year, Department and Category slicers to isolate problem areas.
- Keep the DAX variance sign convention consistent across Excel and Power BI.

20. Final Submission Checklist

Deliverable / Requirement	Status	Notes
Cleaned dataset	Completed	1,061 records, 11 fields.
Cleaning documentation	Completed	Project notes and expanded cleaning log.
Pivot Tables	Completed	Fiscal Year, Division, Function, Fund, Department.
Excel dashboard	Completed	KPI cards and charts documented with screenshots.
Power BI fact/dimension model	Completed / documented	Fact_Budget plus dimension tables.
Relationships	Documented	Validation rules included.
DAX	Documented	Measure dictionary included.
Power BI visualizations	Documented	Page map and visualization catalogue included.
Insights	Completed	Descriptive, diagnostic, predictive and prescriptive sections.
Gap analysis	Completed	Known inconsistencies and documentation gaps addressed.
Final PDF documentation	Completed	This file.

BUDGET VS ACTUAL FINANCIAL ANALYSIS DASHBOA



Final reference image – Excel dashboard.

21. Appendix – Source Screenshots

A. Excel Pivot / Dashboard evidence

By Division	
Variance	Division
1,486,839	EXPENSE
383,977	REVENUE
-2,708,818	

By Fund	
Variance	Fund
-1,240,485	GENERAL FUND
-459,121	TRANSIT FUND
-534,197	DEBT SERVICE FUND
-173,101	STORMWATER MANA
1,442,377	PUBLIC HOUSING OPE
151,614	OFF-STREET PARKING
115,573	CAPITAL IMPROVEME
-140,874	VEHICLE MAINTENANC
298,072	VEHICLE REPLACEMEN
124,112	ON-STREET PARKING
-421,972	AFFORD HSG DEVELOP
	TRANSIT CAPITAL RES
	GRANTS FUND
	DOWNTOWN SERVICE
	LIBRARY GIFT FUND
	COMPUTER REPLACEN
	CAPITAL RESERVE FUN

Figure 15. Pivot variance and dimension summary.

B. Power BI modelling evidence

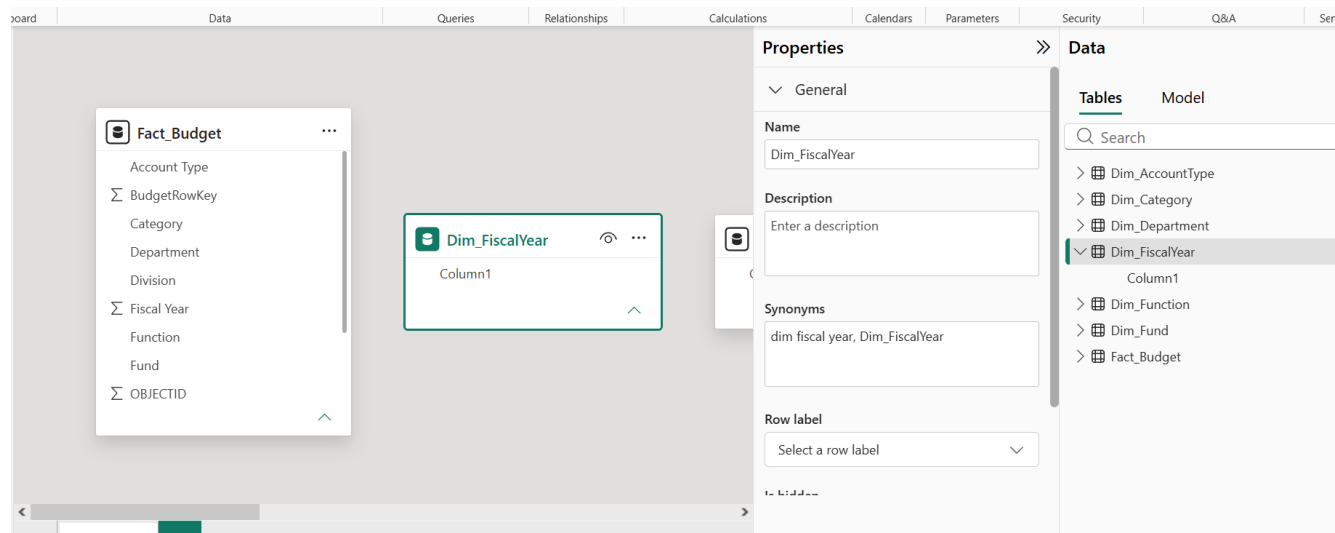


Figure 16. Power BI model screenshot.

C. Power BI relationship evidence

From table

Fact_Budget

Account Type	BudgetRowKey	Category	Department	Division	Fiscal Year	Function
Expense	6	1188389	TRANSIT	EXPENSE	2024	DEBT S
Revenue	27	1359731	TRANSIT	REVENUE	2024	DEBT S
Expense	40	20000	TRANSIT	EXPENSE	2022	TRANS

To table

Dim_Fiscal year

Fiscal Year
null
2022
2023

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

Figure 17. Power BI relationship setup screenshot.